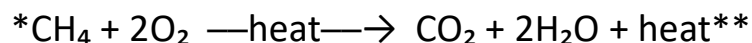
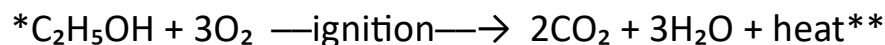


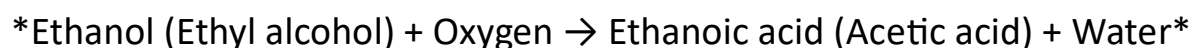
1. COMBUSTION OF METHANE



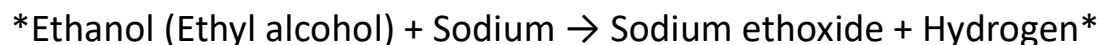
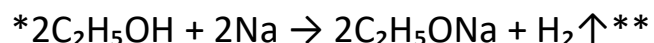
2. COMBUSTION OF ETHANOL



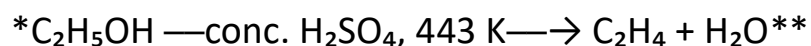
3. OXIDATION OF ETHANOL



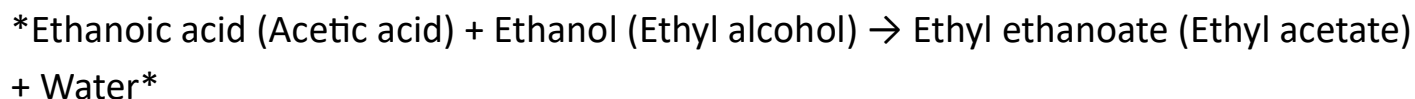
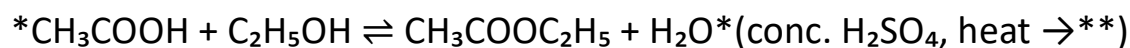
4. ETHANOL + SODIUM



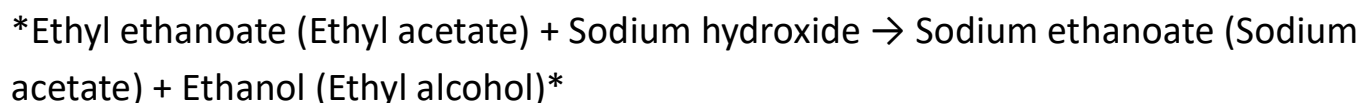
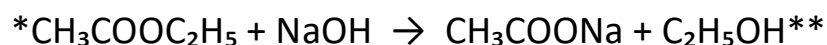
5. DEHYDRATION OF ETHANOL



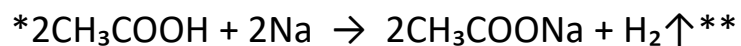
6. ESTERIFICATION



7. SAPONIFICATION



8. ETHANOIC ACID + SODIUM



Ethanoic acid (Acetic acid) + Sodium $\rightarrow$ Sodium ethanoate (Sodium acetate) + Hydrogen

9. ETHANOIC ACID + SODIUM HYDROXIDE



Ethanoic acid (Acetic acid) + Sodium hydroxide $\rightarrow$ Sodium ethanoate (Sodium acetate) + Water

10. ETHANOIC ACID + SODIUM CARBONATE



Ethanoic acid (Acetic acid) + Sodium carbonate $\rightarrow$ Sodium ethanoate (Sodium acetate) + Water + Carbon dioxide

11. ETHANOIC ACID + SODIUM HYDROGEN CARBONATE



Ethanoic acid (Acetic acid) + Sodium hydrogen carbonate (Sodium bicarbonate) $\rightarrow$ Sodium ethanoate (Sodium acetate) + Water + Carbon dioxide

12. HYDROGENATION OF ETHENE



Ethene + Hydrogen $\rightarrow$ Ethane(*Catalyst:** Nickel / Palladium)

13. SUBSTITUTION OF METHANE



Methane + Chlorine $\rightarrow$ Chloromethane (Methyl chloride) + Hydrogen chloride

14. FURTHER CHLORINATION OF METHANE



Chloromethane (Methyl chloride) $\rightarrow$ Dichloromethane (Methylene chloride) + Hydrogen chloride



Dichloromethane (Methylene chloride) $\rightarrow$ Trichloromethane (Chloroform) + Hydrogen chloride



Trichloromethane (Chloroform) $\rightarrow$ Tetrachloromethane (Carbon tetrachloride) + Hydrogen chloride

★ CONDITIONS TO REMEMBER

Heat: Combustion

$\text{KMnO}_4 / \text{K}_2\text{Cr}_2\text{O}_7$: Oxidation

conc. $\text{H}_2\text{SO}_4 + 443 \text{ K}$: Dehydration

conc. $\text{H}_2\text{SO}_4 + \text{heat}$: Esterification

Ni / Pd : Hydrogenation

Sunlight: Chlorination

1. Alcohol — Functional group: $-\text{OH}$

Carbon atoms	IUPAC name	Common name	Molecular formula	
1	Methanol	Methyl alcohol	CH_3OH	
2	Ethanol	Ethyl alcohol	$\text{C}_2\text{H}_5\text{OH}$	
3	Propan-1-ol	Propyl alcohol	$\text{C}_3\text{H}_7\text{OH}$	
4	Butan-1-ol	Butyl alcohol	$\text{C}_4\text{H}_9\text{OH}$	
5	Pentan-1-ol	Amyl alcohol	$\text{C}_5\text{H}_{11}\text{OH}$	formula: $\text{C}_n\text{H}_{2n+1}\text{OH}$
6	Hexan-1-ol	Hexyl alcohol	$\text{C}_6\text{H}_{13}\text{OH}$	

2. Aldehyde — Functional group: $-\text{CHO}$

No.	Molecular Formula	IUPAC Name	Common Name
1	HCHO	Methanal	Formaldehyde
2	CH_3CHO	Ethanal	Acetaldehyde
3	$\text{C}_2\text{H}_5\text{CHO}$	Propanal	Propionaldehyde
4	$\text{C}_3\text{H}_7\text{CHO}$	Butanal	Butyraldehyde